VENUS PROBE PROJECT RULES

v2.2

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2 Introduction

Understanding volcanism on Venus is essential for understanding the planet's geologic activity, heat transfer, and atmospheric evolution. Volcanic outgassing plays a central role in sustaining Venus's dense, CO_2-rich atmosphere, and evidence of present-day volcanism would reveal much about the planet's interior processes and thermal history. Yet, volcanism on Venus is difficult to detect remotely: spacecraft in orbit can only infer indirect signs, while landers have a very limited lifetime, caused by the extreme heat and pressure at the surface, preventing long-term observations.

A proposed solution is to exploit the short time window available as a probe falls through the atmosphere. During descent, sensitive instruments can attempt to capture the subtle signatures of explosive activity. On Venus, low-frequency acoustic waves propagate especially well through the dense atmosphere, traveling vast distances with little attenuation. By equipping a probe with sensitive microphones, it becomes possible to detect these signals, separating them from background atmospheric noise.

Such measurements could provide direct, ground-truth evidence of active volcanism on Venus, complementing orbital data and advancing our understanding of planetary processes under extreme conditions. Exploring these methods in an analogue mission on Earth offers a unique opportunity to test technologies and strategies that could one day be applied to real planetary exploration.

3 Mission Overview

This project simulates an analogue Venus mission, aiming to develop technologies and concepts that could one day be applied inside the Venus atmosphere. Students will work in teams of 6 - 7 people to design and build a small probe, which will be released from a light aircraft at an altitude of 600 meters. Each team will receive a kit containing electronic components along with a software framework to get started, plus a budget for 3D-Printing and additional material.

4 Mission Objectives

The scientific objectives are to **estimate the detonation rate and the detonation time** of the Vulcan-simulating explosives using acoustic measurements, while falling towards the ground. As the link budget from Venus to Earth is very small, data must be processed and analysed on-board, and only the results can be sent back to earth. The larger Pioneer probe, for example, had a maximum data rate of 256 bits/s[1]. To mimic this small link budget, Long Range (LoRa) radio transmission will be used in this mission. The spacecraft must also be able to receive commands.

Missions to Venus are rare, so scientists will want to add additional scientific payloads to the mission. It is the task of each team to come up with at least two science cases and define the scientific objective and requirements for those two payloads. The instruments that are already on the probe may be used for this science cases, but additional instruments can be added.

To locate the source of the detonation, scientists have proposed a pyramid-shaped microphone array. By comparing the differences in the arrival time of the acoustic wavefront at each microphone, the array can be used to reconstruct the direction and approximate distance of the explosion, given that the microphones are spaced sufficiently far apart. Implementing such a system is complex and requires significant effort, so it will be considered an optional task and will not be graded — though it is a task that presents a valuable engineering opportunity.

[1] G. J. Nothwang, "Pioneer Venus Spacecraft Design And Operation," in IEEE Transactions on Geoscience and Remote Sensing, vol. GE-18, no. 1, pp. 5-10, Jan. 1980, doi: 10.1109/TGRS.1980.350251

5 Provided Material

Each team will receive a kit containing electrical components for use in the system. These components **include pre-soldered header pins, which shall not be removed**. However, the header pins may be soldered onto a stripboard or similar installation if needed. **At the end of the project, the kit must be returned to the lecturers in its original condition. Please refer to the hardware documentation for more Information.**

Each team will be provided with materials to construct a parachute, including 5 square meters of parachute fabric and 15 meters of parachute lines. If you prefer alternative materials or a different design for drag reduction, you are welcome to implement these changes. However, the cost of these materials must be deducted from the budget.

6 Budget

In addition to the electronics kit, the teams are provided a **budget of 50 CHF**. This money **must be used for all expenses**, but also to trade with other teams.

Teams are allowed and encouraged to buy and sell their solutions to other teams – parachutes, software, etc. The prices and details of their "contracts" are to be agreed upon based on individual deals between teams and must be documented. The TAs & Lecturers act as a court of arbitration and have the final say.

Note that it is **not allowed to sell/buy or share data, methods, or other contents relevant for the System Performance Report** to/from other teams before the reports are submitted. These reports must be written individually.

Teams are also **not allowed to freely 3D print prototypes** at home without deducting all cost from their budget. Cost is CHF 0.02 per gram of filament. However, each team will get **1 roll of PLA filament for free**.

7 Project Timeline

03.12.2025:	**Preliminary Design Review** (Location: HG E 41)
17.12.2025:	**Critical Design Review** (Location: Dübendorf)
19.12.2025:	**Data Package Submission**
21.01.2026:	**Test Reports Submission**
23.01.2026:	**Flight Readiness Review** (Location: Dübendorf)
28.01.2026:	**Start of Launch Window** (until 04.02.2026)

8 Grading

The Project is graded. The total grade is calculated based on the following table:

Table 1: Grading

Item	Grade Weight
Data Package	**40 %**
Test Plan	20 %
System Performance Report	20 %
Critical Design Review – Presentation	**20 %**
Test Reports	**20 %**
System Performance	**20 %**

The System Performance is graded based on an internal benchmark. This may include functionality, engineering and design choices and manufacturing quality (list not exhaustive).

9 Preliminary Design Review – Short Presentation (not graded)

The PDR Short Presentation takes place on **03.12.2025, starting 13:15 in HG E 41** and is not graded. It is a good opportunity to receive feedback on the design before the CDR Presentation.

The review is done in a 10-minute presentation, followed by a 10-minute feedback session. The following points shall be addressed in the presentation:

- **System:** The audience shall understand the system in as much detail as necessary to give valuable feedback. Use the learned System Engineering Tools to achieve so (**Preliminary Requirements, Product Tree, Concept of Operation, etc.**).
- **Secondary Payloads:** Explain Secondary Payloads and their science cases
- **Virtual Prototype:** The Teams should present a “baseline prototype”, consisting of the building plans for a first iteration of the final system. All components shall be baselined, but nothing build yet.

10 Critical Design Review – Presentation (graded)

The CDR is a **15 min graded presentation** where all members of the team must present. The presentation takes place on **17.12.2025 in Dübendorf, starting 9:15**, followed by 5 min Q&A and an individual feedback session in the afternoon. The teams must **upload their presentation to Moodle by 16.12.2025, 23:59**.

The CDR shall include the following points:

1. **System Design:** The final system design shall be presented. The design is finished with no further major changes expected.
2. **System Engineering:** Tools learned in the course shall be used to explain the system. This can include a product tree, concept of operations, functional breakdowns or similar methods.
3. **Requirements:** The three most relevant requirements shall be explained.

Graded are both presentation style and content.

11 Data Package (graded)

The Data Package is graded and consists of two separate reports: The **Test Plan** and the **System Performance Report**. It must be submitted once per group on Moodle until **19.12.2025, 23:59.**

11.1 Test Plan

The Test Plan must be submitted as a separate document. The focus should be on testing – how and which requirement is tested and at what time. It shall include **Failure Mode & Effects Analysis** and the **V-Model**, as learned in the lecture.

Please be aware that the Test Plan **will be graded very strictly** and test reports of tests not in the test plan will not be accepted. Grading aspects will be completeness and feasibility.

11.2 System Performance Report

The students shall predict different performance characteristics of their system and of themselves as a team:

1. **Hardware Calculations:** System mass, system dimensions and centre of gravity
2. **Data Budget:** Amount of data generated, stored, and downlinked, down- and uplink budget
3. **Work Package Breakdown:** Working hours per package, milestones and delivery dates, schedule, cash budget
4. **List of Requirements and Science Matrix:** A detailed list of all requirements as learned in the course. For each secondary payload, a science case shall be presented, and the requirements for the payload shall be specified.
   The Science Matrix should consist of the following items:
   a. Descend time
   b. Horizontal drift
   c. Signal-to-Noise Ratio
   d. Detection uncertainty

The numerical values will be noted and compared to the actual values a launch and during descent. The process and documentation will consist of the grade. The difference between measured and predicted data will be used as one of the scoring metrics for the competition.

It is proposed that the team split into groups of 1-2 ppl, with each group picking one characteristic. Each group shall submit a 3–5-page report. A document of comparable size (such as a 2-page report accompanied by a well-commented Excel file) is also acceptable.

Each report is graded individually, so make sure to name the authors. It is also possible to submit the whole System Performance Report (15-20 pages) as a group.

12 Test Reports (graded)

The three most relevant test reports are graded. **One of them must be the safety and system handling test report**. They are graded for completeness and application of testing best practices. The test reports must outline:

- Test objectives
- Test procedure
- Test results

The results need to be put into context for the entire system. Each test report shall be about 3-5 pages long. They must be submitted once per group on **Moodle until 21.01.2026, 23:59**.

13 Flight Readiness Review – Presentation (not graded)

The main purpose of the FRR is to ensure that the launch of the system is safe. Therefore, a key focus will be on the **Parachute Release Mechanism and the System Handling**. The group must show that the Safety Requirements are fulfilled. Otherwise, the system will not pass the FRR.

If the system passes the FRR, the launch provider will collect the system after the presentation and store them until launch. Any further access to the system after the FRR is not allowed and not possible. This **requires the creation of an instruction manual** for the launch provider, which is handed in at the FRR. The instruction manual shall tell the launch provider how to prepare the system before the drop. These preparation steps shall take no longer than 5 minutes. It is possible to include a GPS acquisition step before the system gets transferred to the airplane (for timekeeping and to get faster satellite-lock).

14 Launch

The systems are flown up to an altitude and dropped by a third-party provider. They will throw the systems out of the airplane in flight.

14.1 Minimum Meteorological Conditions

Minimum meteorological conditions	
Ceiling (defined as altitude of 4/8 cloud cover)	2500 ft AGL
Wind speed (max.)	7 kts
Gusts (max.)	5 kts
Minimum visibility	5 km

14.2 Drop Altitude

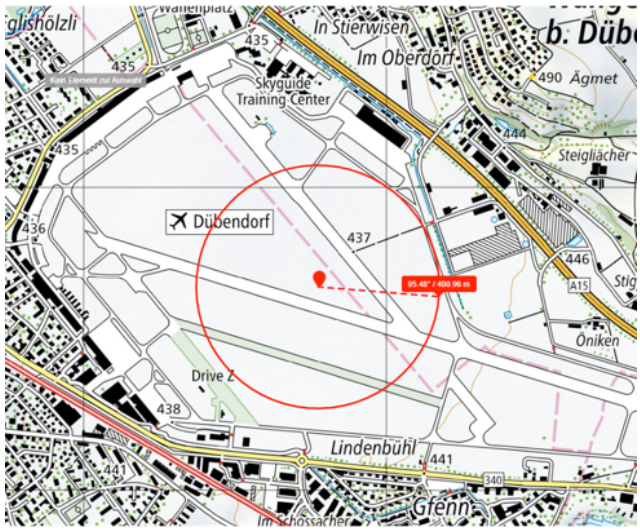
The drop altitude is 2000 ft AGL. In case of constantly low ceiling (<2500 ft AGL), it can be lowered to 1000 ft AGL. The decision will be made by the launch provider.

14.3 Drop Procedure

1. The system power button (or switch) is pressed / switched before engine start-up, up to 30 minutes before drop.
2. If the status LED / display does not indicate a successful start-up, the launch provided will press a reset button, which might reset the system to an operational state.
3. There will be no further interactions with the system. The team can include a single “drop” button, which will be pressed approx. 1 minute before the drop, if desired.
4. The system is not **allowed to take any measurements once landed on the ground.**

14.4 Drop Location

The drop will be done at the indicated location in the map. If wind speed as reported by Dübendorf tower are above 2 kts, the aircraft operator will correct the drop location based on the current ground windspeeds as reported by METAR and on drop altitude by drift.



15 Safety Requirements

These rules ensure safe operations onboard the aircraft and on the ground. Compliance will be checked during the FRR and needs to be demonstrated by tests.

15.1.1 Size & Mass Requirements

R1.1 The system in its launch configuration shall be fully contained within a rectangular spatial envelope where:

- The total calculated volume of the envelope does not exceed 16’000 cm^3.
- No single dimension (Length, Width, or Height) exceeds 40 cm.

R1.2 Loose cables or strings are not allowed.

R1.3 The system shall be no heavier than 1000 g

R 1.4 The system shall be no lighter than 100 g

15.1.2 User Interface

R2.1 The system shall be possible to operate with thin gloves

R2.2 The system shall have a Power-On button to be pressed before engine start-up on ground, approximately 30 minutes before drop.

R.2.3 The system shall have a Status LED and/or a screen indicating the current operational state of the system.

R2.4 The system shall have a Reset Button / Reset Procedure to press in case the operation state is not “READY”

R2.5 The interfaces shall be clearly labelled

15.1.3 Drag Modification Devices

R3.1 The system shall have a drag modification device which limits the terminal velocity to no faster than 10 m/s.

R3.2 If a parachute and/or other deployable devices are used, they shall open no sooner than one second after the drop.

15.1.4 Dangerous Devices

R4.1 No pyrotechnical, chemical, or explosive devices shall be installed on the system.

R4.2 Mechanisms using heat generation (e.g. nichrome wire, heating coils or resistive elements) are not permitted.

16 Data & Transmission Rules

In each electronics kit, there is a LoRa transmission module with a corresponding antenna. To align to rules and regulation, and for interoperability with the ground station, the following settings must be used:

Parameter	Range
Frequency	868.30 MHz
Transmit Power	Max. 14 dBm EIRP (25 mW radiated power)
Duty Cycle	Max. 1%
Coding Rate	4 / 6
Spreading Factor	8
Bandwidth	250 kHz

During the descent, data should be transmitted to a ground station, which will be operated by the TA team. As the LoRa modules can only work on one frequency at a given time, two radios sending data at the same time would lead to congestion and, possibly, loss of data. Therefore, the ground station will periodically request data downlink from each probe. It will request scientific data as well as telemetry data. To ensure compatibility, please implement the communication protocol outlined below. For backup reasons, the teams should **store all data generated on the on-board SD card**.

16.1 Transmission Rules

1. Teams are **not allowed to send any commands** to the probes during the competition.
2. The probes are **not allowed to send the Command Packet.**
3. The probes shall **only send packets if commanded to do so** by the base station.
4. Packets with **not-specified packet ID's or packet formats will be dropped.**

16.2 Communication Protocol

There are three different types of packets, as visible in Table 2. The ground station will periodically send the command packet to the probes. The ground station will send a command packet a new probe approximately every 3 seconds, which means each probe, with margin, is expected to be commanded to send data every 20 seconds. Once the Command Packet is received at the target probe, it shall respond no sooner than 50ms but no later than 1.5s after the packet is received. All transmission from the probe shall be finished within 2s after receiving the Command packet. The probe shall respond with one of the two packets 0x1X or 0xAX, depending on the command of the ground station.

Table 2: Packet Overview

Packet ID	Name	Purpose	Size [byte]	Description
0x00	CMD	Command Packet	2	Command packet sent from the base station.
0x1X	SCI	Scientific Data	TBD	Data packet containing scientific data
0xAX	TEL	Telemetry Data	16	Data packet containing telemetry data.

16.2.1 Command Packet

The Command Packet will be sent only from the base station. The CMD entry has two binary flags (1 = True, 0 = False) that indicate what the base station requests. It also includes the team number encoded in binary. For details refer to Table 4.

Table 3: Command Packet Structure

Byte Index	Content	Description
0	0x00	Command packet indicator byte
1	CMD	CMD Byte

Table 4: Command Byte structure

CMD [7]	CMD [6]	CMD [5]	CMD [4]	CMD [3:0]
RESERVED	**RESERVED**	TELEMETRY DATA	SCIENTIFIC DATA	TEAM ID

16.2.2 Scientific Data

Byte 0 has two symbols in hexadecimal system. The first symbol is 0x1, the second symbol is the Team ID encoded in binary. The Detonation Localisation and Telemetry Data fields are optional for this packet, while the Detonation Event and Secondary Payload fields must be present. You are completely free to design the content of these; just make sure it matches exactly the number of bytes. Use zero-padding if you are not using all bytes.

Table 5: Scientific Data Packet Structure

Size [bytes]	Content	Description
0	0x1X	Packet ID [7:4] and Team ID [3:0]
4	Timestamp	4-byte absolute time (Unix epoch seconds)
1	Bitmap	Data indicator byte
20	Detonation Event	20-byte detonation event data
20	Secondary Payload	20-byte secondary payload data
25	Detonation Localisation	Optional 20-byte detonation localisation data
10	Telemetry Data	Optional 10-byte telemetry data
1	CRC	1 byte CRC-8-CCITT checksum

The bitmap field indicates which data is present in the packet, if the indicator bit is set to 0, then the data is not included, if it is set to 1, then the data is included. **Make sure to always set these to correctly match your packet structure, else the ground station will not be able to parse it, and the packet will be dropped.**

Table 6: Bitmap structure

[7:4]	[3]	[2]	[1]	[0]
RESERVED	TEL DAT	DET LOC	SEC PLD	DET EVT

16.2.3 Telemetry Data

Byte 0 has two symbols in hexadecimal system. The first symbol is 0xA, the second symbol is the Team ID encoded in binary.

Table 7: Telemetry Data Packet Structure

Byte Index	Content	Description
0	0xAX	Packet ID [7:4] and Team ID [3:0]
1-3	Latitude	Signed 24-bit integer, degrees, scaled by 10^5
4-6	Longitude	Signed 24-bit integer, degrees, scaled by 10^5
7-8	Velocity	Signed 16-bit vertical velocity, cm/s (resolution of 0.01 m/s)
9-10	Altitude	Unsigned 16-bit integer, meters x 10, relative to sea level
11-14	Timestamp	4-byte absolute time (Unix epoch seconds)
15	CRC8	1 byte CRC-8-CCITT checksum

17 Testing Days

These test days are provided by the TA team, as the following tests are quite complex. They will only take place at the announced dates and will not be repeated. **Please come to the tests prepared and on time**.

17.1 Parachute Inflation and Load Test

Starting from **12.01.2026** there will be a Parachute Inflation and Load Test in **Birrfeld**. As this airport is active, we can't ensure a fixed date yet. We will try to notify you as early as possible on Moodle. This test will verify that the parachute inflates and can sustain the opening load. **This will not be a parachute deployment mechanism test** (this test must be conducted by each group individually). The parachute will be mounted on a car, accelerated to 100 km/h and then released into the airstream. For additional information please wait for the Moodle post.

17.2 Ground Station Connectivity Test

Starting from **12.01.2026 in Dübendorf**, there will be a LoRa ground station connectivity test. This test will verify that the system can communicate with the ground station. The exact date will be published on Moodle as early as possible. For additional information please wait for the Moodle post.

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